Exercise 3: SQL Case Statements

Question:

1. Classify each product by price
- 'Expensive' if price > 1000
- 'Mid range' if price between 100 and 1000
- 'Budget' if price < 100

Expected Output Columns
- product_name
- Price
- Price category

```
SELECT product_name,
       Price,

CASE
    WHEN price > 1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000 THEN 'mid-range'
    ELSE 'Budget'
END AS price_category
FROM products;
```

Product_name	Price	Price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2. Question:

Label each order
- 'High value' for orders ≥ 1000
- 'Medium value' for 500 - 99.99

'Low value' for orders < 500

Expected outcome colums

- Customer-name
- amount
- Order-value-category

SQL Syntax:

```
SELECT customer name,
       amount,
  CASE
  WHEN amount ≥ 1000 THEN 'High value'
  WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium
  ELSE 'Low value'
  END AS order_value_category
  FROM Orders;
```

Result Table

Customer-name	amount	Order-value-category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200.00	High value

3. Question:

Categorize employee position:

- If in 'IT' and Salary > 80000 → 'Senior IT'
- If in 'HR' and Salary > 55000 → 'Experienced HR'
- Otherwise → Staff

Expected outcome colums

- emp-name
- department
- Salary
- position-level

SQL Syntax

```
SELECT emp-name,
       department,
       Salary,
CASE
WHEN department = 'IT' AND salary>80000 THEN 'Senior IT'
WHEN department = 'HR' AND salary>55000 THEN 'Experienced HR'
ELSE 'Staff'
END AS position-level
FROM employees;
```

emp-name	department	Salary	position-level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Question

Assign a letter grade:

- ≥ 90: 'A'
- 80-89: 'B'
- 70-79: 'C'
- 60-69: 'D'
- <60: 'F'

Expected Output columns:

- Student-name
- Score
- grade

```
SELECT student-name,
       Score
  CASE
  WHEN Score >= 90 THEN 'A'
  WHEN Score BETWEEN 80 AND 89 THEN
  WHEN Score BETWEEN 70 AND 79 THEN
```

WHEN Score BETWEEN 60 AND 69 THEN 'D'
ELSE 'F'
END AS grade
FROM Students;

Result table.

Student_name	Score	grade
Ama	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	63	D

5. Question

Label delivery performance:

- ≤ 30 mins: 'Fast'
- 31–60 mins: 'On Time'
- 60 mins: 'Late'

Expected output columns

- delivery_id
- delivery_time_minutes
- performance

SELECT delivery_id,
delivery_time_minutes
CASE
WHEN delivery_time_minutes ≤ 30 THEN 'Fast'
WHEN 'delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
ELSE 'Late'
END AS performance
FROM deliveries;

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	fast
4	65	Late
5	100	late

6. Question:

Convert priority ~~labels~~ to labels:

- 3 → 'High'
- 2 → 'medium'
- 1 → 'Low'

Expected Outcome Colums

- Issue_type
- Priority
- Priority_label

SQL Syntax

SELECT

CASE
WHEN priority = 3 Then 'High'
WHEN priority = 2 Then 'Medium'
WHEN priority = 1 Then 'Low'
END AS priority_label
FROM tickets;

Result table:

issue_type	Priority	Priority_label
Login issue	1	Low
Serverdown	3	High
Slow system	2	medium
Email error	2	medium
Password reset	1	Low

7. Question:

Calculate attendance % and classify:

- ≥ 90% - 'Excellent'
- 75-89% - Good
- 75% - Needs improvement'

Expected Output columns:

- Student_id
- attendance_percentage
- attendance_status

SQL Syntax:

```
SELECT student_id,
    (days_present * 100.0/total_days) AS attendance_pe
    CASE
WHEN (days_present * 100.0/total_days) ≥ 90 THEN Excellent
WHEN (days_present * 100.0 / total_days ) BETWEEN 75 AND 89 Good
ELSE 'Needs Improvement'
END AS Attendance_Status
FROM attendance
```

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Student_ID	attendance_percentage	attendance_Status
1	90.00	Excellent
2	60.00	Needs Improvement
3	96.00	Excellent
4	50.00	Needs Improvem
5	100.00	Excellent

8. Question:

Label Stock Status:

- 0 → out of stock
- 1-5 → Low stock
- 5 → In stock

Expected output colums

- Product_id
- Stock_qty
- Stock_status

```
SELECT
        product_id,
        Stock_qty,
  CASE
WHEN Stock_qty = 0 THEN 'OUT OF STOCK'
WHEN Stock_qty BETWEEN 1 AND 5 THEN 'LOW STOCK'
 ELSE 'In Stock'
 END AS Stock_Status
FROM product_inventory;
```

Result table

product_id	Stock_qty	Stock_Status
1	5	In Stock
2	0	out of stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

9. Classify by size

- ≥ 25 → 'Large'
- 10 - 24 → 'Medium'
- < 10 → Small

Expected output colums

- subject
- enrolled_students
- Class_size_category

SELECT

Subject

enrolled_students

CASE

WHEN enrolled_student >= 25 THEN 'Large'

WHEN enrolled_students BETWEEN 10 AND 24 THEN 'medium'

ELSE 'Small'

END AS Class_size_category

FROM Classes,

Subject	enrolled_students	Class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Question:

Apply discount Flag:

- If payment_method = 'Cash' and amount ≥ 200 → Eligible (Discount)
- Otherwise → Not eligible

Expected Output Columns

- Payment_Id
- Payment_method
- amount
- discount_eligibility

SQL Syntax

SELECT Amount,

Payment_Id,

Payment_method

CASE

WHEN payment_method = 'cash' AND amount ≥ 2000 THEN 'Eligible for Discount'
ELSE NOT Eligible
END AS discount_eligibility
FROM Payments

Payment_id	Payment_method	amount	discount elig
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for discou
3	card	150.00	Not eligible
4	Paypal	75.00	Not eligible
5	Cash	300.00	Eligible